

Classical Mechanics

Lecture 2: Forces, Phase Space, and Conservation Laws

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Chapter 1

Forces, Phase Space, and Conservation Laws

Overview. A law of motion must do more than relate familiar physical quantities. It must specify the state of a system, carry that state from one instant to the next, and reveal which distinctions and quantities survive the motion. We shall reach Newtonian mechanics by first testing a plausible but wrong force law, then enlarge the state from position to phase space, and finally derive conservation of momentum and energy.

1.1 A Useful Wrong Law

Everyday motion is dominated by friction. Push a heavy object across a floor and it moves; stop pushing and it soon stops. A larger push generally produces a larger speed, while a heavier object requires a harder push. In such a world, it is natural to suppose that force produces velocity itself. This was the basic intuition behind the Aristotelian description of motion.

Ice skating supplies the missing experience. A skater needs force to start, stop, or change direction, but not to continue gliding once in motion. Had Aristotle worked on nearly frictionless ice rather than ordinary ground, the defect in the force–velocity intuition would have been much harder to miss.

To expose what is wrong with that intuition, let us give it a precise modern form. We write the deliberately incorrect equation

$$\mathbf{F} = m\mathbf{v}. \tag{1.1}$$

This is not Newton’s law and should not be memorized as physics. Its value is diagnostic: once a verbal rule has become an equation, we can ask whether it predicts motion and whether it preserves enough information to reconstruct the past. The brief classroom discussion of the Greek word for force points to *dynamis*, the root also heard in words such as dynamo and dynamics, but the terminology is secondary to the test of the law itself.

Restrict the motion to one spatial coordinate. Since $v = \dot{x}$, the law is

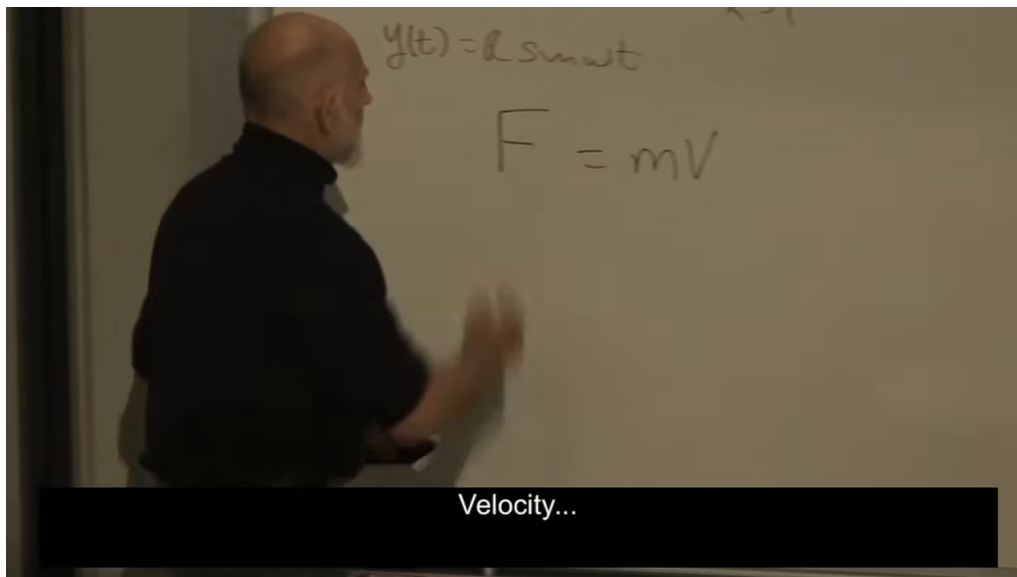


Figure 1.1: The intentionally wrong force–velocity law used to test predictability and reversibility.

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$$F(t) = m\dot{x}(t). \quad (1.2)$$

Now divide time into short intervals of length δ . Over one interval,

$$\dot{x}(t) \simeq \frac{x(t + \delta) - x(t)}{\delta}. \quad (1.3)$$

Substitution into Eq. (1.2) gives

$$x(t + \delta) = x(t) + \frac{\delta}{m}F(t). \quad (1.4)$$

This has the form of a genuine update rule. If the present position and the force are known, the position at the next stroboscopic instant follows. When the force vanishes, however, Aristotle’s rule gives $\dot{x} = 0$: the object must remain at rest. Motion without a sustaining force is excluded from the start.

1.1.1 A spring in Aristotle’s mechanics

Consider a particle attached to a spring, with equilibrium at $x = 0$. A displacement produces the restoring force

$$F(x) = -kx, \quad k > 0. \quad (1.5)$$

Inserting this force into Eq. (1.4) yields

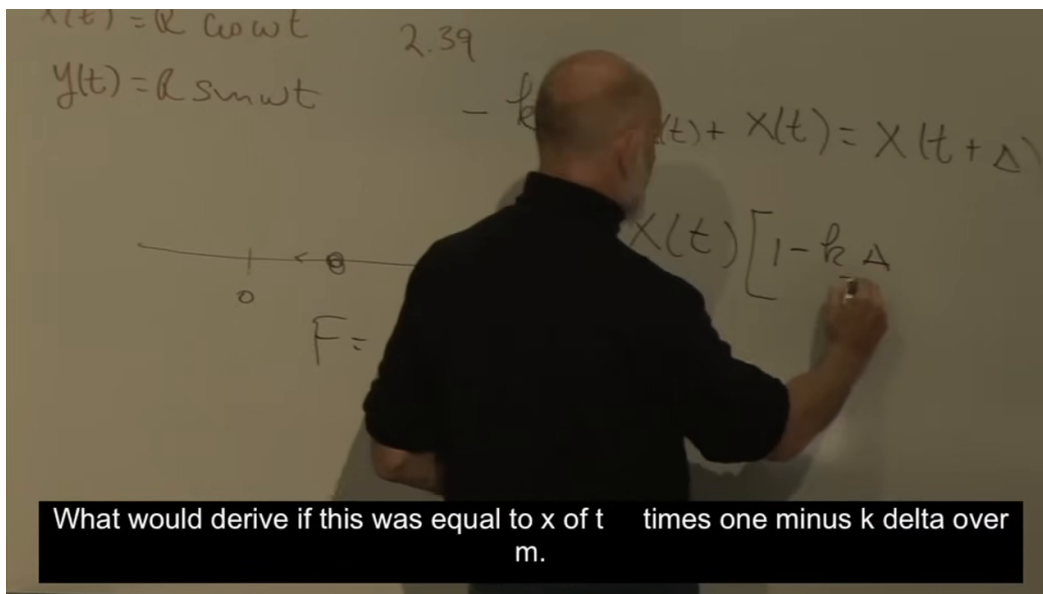


Figure 1.2: The discrete Aristotle update with a restoring force. Each step multiplies the displacement by a factor slightly smaller than one.

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$$x(t + \delta) = \left(1 - \frac{k\delta}{m}\right) x(t). \quad (1.6)$$

For sufficiently small δ , the factor in parentheses lies just below one. Every update therefore reduces the displacement by the same fraction:

$$x_n = \left(1 - \frac{k\delta}{m}\right)^n x_0. \quad (1.7)$$

The continuous equation and its solution are

$$m\dot{x} = -kx, \quad (1.8)$$

$$x(t) = x(0)e^{-kt/m}. \quad (1.9)$$

The spring does not oscillate. Released from any displacement, it approaches the origin exponentially, moving more and more slowly as it comes in.

The future is easy to predict: every initial displacement approaches the same resting configuration. The past is another matter. Suppose the particle is observed very near the origin with only finite accuracy. It could have begun at any of many distinct displacements, because all of their exponentially small remnants eventually lie inside the same measurement tolerance. With perfect mathematical precision Eq. (1.9) can be run backward, but practical distinctions between histories are steadily erased. The law is therefore suspicious from the standpoint of reversible fundamental dynamics.

This does not mean that Aristotle was irrational. The friction-dominated motion he described really is dissipative. The mistake was to promote that effective behavior to a fundamental law. Friction

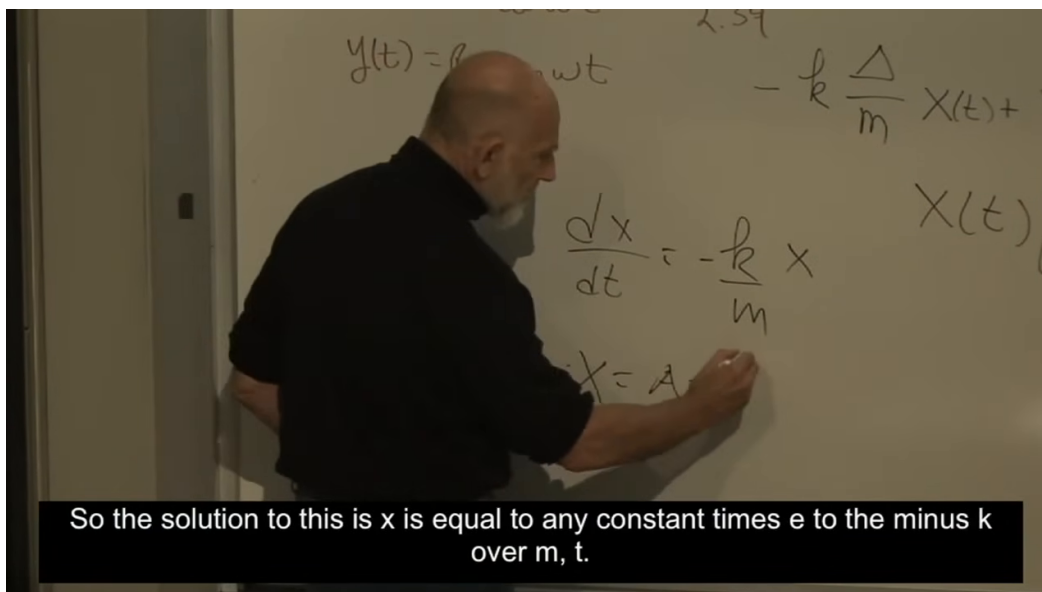


Figure 1.3: The exponential solution of the Aristotelian spring equation. The displacement decays rather than oscillates.

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is itself a force arising from complicated microscopic interactions. When a hand pushes an object at roughly steady speed, the applied force is balancing friction; the applied force is not creating velocity as such.

1.2 Newton's Law and Its Physical Meaning

The Newtonian correction is that force changes motion:

$$\mathbf{F} = m\mathbf{a} = m\ddot{\mathbf{x}}. \quad (1.10)$$

Before treating this as an evolution equation, two apparent circularities must be untangled. The first concerns frames of reference. One often hears that Newton's laws hold in inertial frames and that an inertial frame is one in which Newton's laws hold. The physical assertion is not that circular definition. It is the empirical claim that there *exist* frames, described by suitable rods and clocks, in which Newton's laws take their simple form. Frames accelerating or rotating relative to them require additional apparent forces. How the equations transform in those other frames can be worked out later.

The second circularity concerns force and mass. If force is defined as mass times acceleration and mass as force divided by acceleration, the equation seems empty. An operational experiment separates the definitions from the law. Compare an object with a standard kilogram by balancing the two, then attach the object to a spring balance in otherwise force-free space. Define one unit of force by a fixed small extension of the spring. Two identical springs, each stretched by that same amount, provide two units of force and produce twice the acceleration. It is safer to add a second spring than to stretch one spring twice as far, because a real spring eventually becomes nonlinear or

breaks. Keeping one spring but joining two equal masses produces half the acceleration; three equal masses produce one third.



Figure 1.4: The spring-balance thought experiment used to define equal units of force and test acceleration proportional to force.

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These comparisons give empirical content to

$$a \propto F, \quad a \propto \frac{1}{m}, \quad F = ma. \quad (1.11)$$

The word *particle* will be used for any small object whose internal structure is irrelevant, not only for an elementary particle.

1.2.1 Why position alone is not a state

We now test Newton's equation by the same stroboscopic method. The centered finite difference for acceleration is

$$\ddot{x}(t) \simeq \frac{x(t + \delta) - 2x(t) + x(t - \delta)}{\delta^2}. \quad (1.12)$$

Thus Newton's equation implies

$$x(t + \delta) = 2x(t) - x(t - \delta) + \frac{\delta^2}{m}F(t). \quad (1.13)$$

The next position cannot be obtained from $x(t)$ alone. We also need the previous position $x(t - \delta)$, or equivalently the velocity at the present instant. At first this makes Newton's equation appear less predictive than Aristotle's. The correct conclusion is instead that position alone was an incomplete

description of the state. To know everything relevant at one instant means to know enough data to determine the subsequent motion. For a Newtonian particle those data are position and velocity.

1.3 Momentum and Phase Space

Introduce momentum by

$$\mathbf{p} = m\mathbf{v} = m\dot{\mathbf{x}}. \quad (1.14)$$

The mass of a fixed Newtonian system is taken to be constant. An object can lose a piece, as a pen can lose its cap, but then the object being described has changed; the chosen closed system still has constant total mass. With constant m , Newton's second law becomes

$$\mathbf{F} = \dot{\mathbf{p}}. \quad (1.15)$$

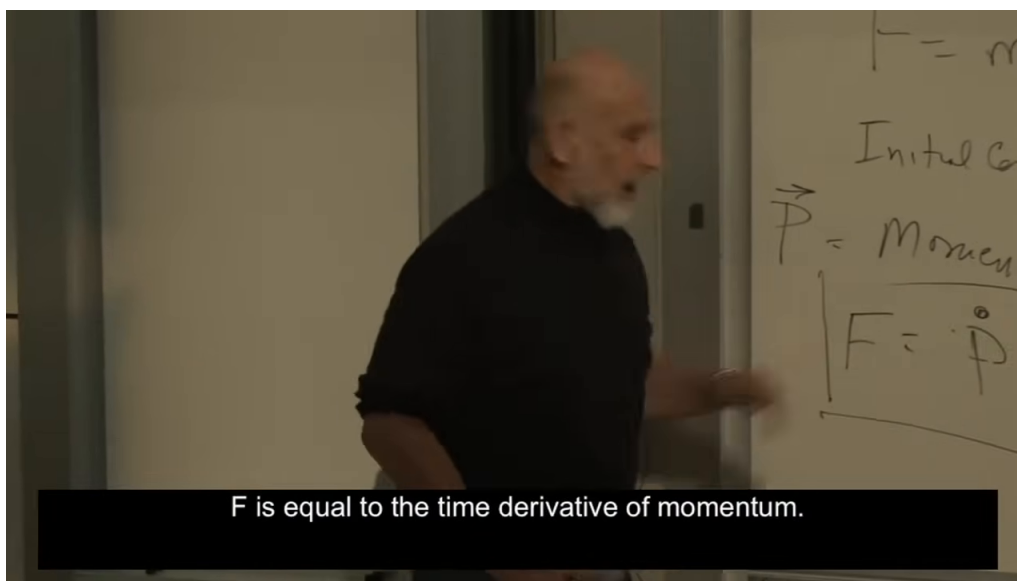


Figure 1.5: Newton's law written as the time derivative of momentum.

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For one coordinate, Newtonian motion can now be written as two first-order equations for two independent state variables:

$$\dot{p} = F(x, t), \quad (1.16)$$

$$\dot{x} = \frac{p}{m}. \quad (1.17)$$

One equation says how the force changes momentum; the other says how momentum changes position. Their first-order discrete form is

$$p(t + \delta) = p(t) + \delta F(x(t), t) + O(\delta^2), \quad (1.18)$$

$$x(t + \delta) = x(t) + \frac{\delta}{m} p(t) + O(\delta^2). \quad (1.19)$$

Given $x(t)$ and $p(t)$, these equations determine the next pair, which then becomes the input for the next step. The trajectory is built by repeated updates.

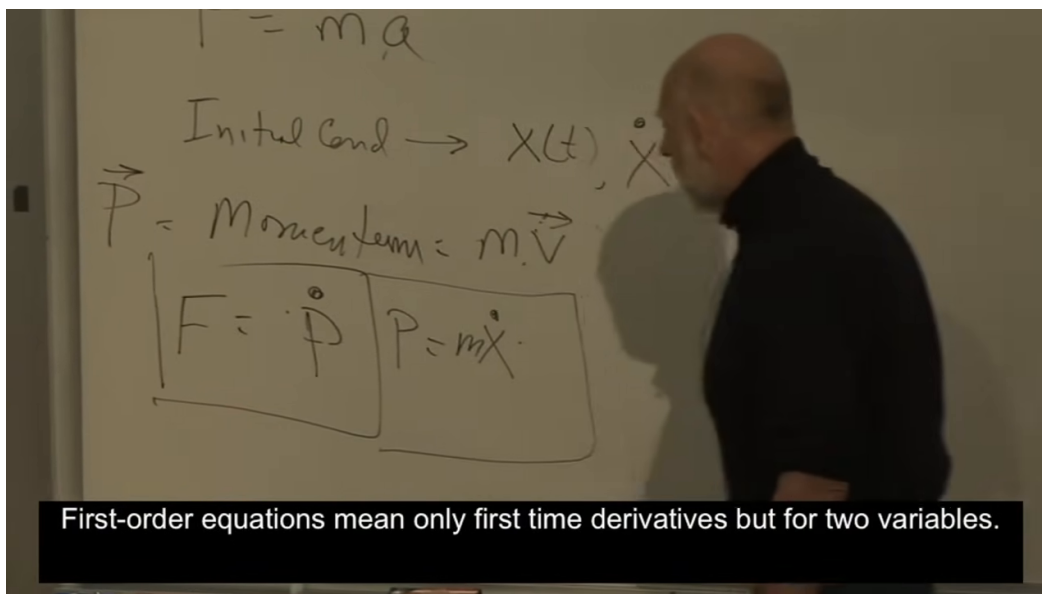


Figure 1.6: The two first-order equations and the corresponding initial data. Position and momentum together specify the state.

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The space of all pairs (x, p) is called *phase space*. For one particle moving in three dimensions it has six coordinates,

$$(x, y, z, p_x, p_y, p_z). \quad (1.20)$$

Phase space is the space of complete initial conditions. Configuration space records positions only; phase space records precisely the data needed by the law of motion.

Question from the audience. If the force is a function of position rather than a known function of time, does the update become more complicated?^a

Response. The equations remain predictive. Once x and p are known, the value $F(x)$ is known. Equation (1.16) advances p , Eq. (1.17) advances x , and the updated pair can be used again.

^aLecture timestamp: 00:36:27–00:37:07.

1.4 The Newtonian Spring

Return to the spring so that the two laws can be compared on identical physical ground. Newton gives

$$m\ddot{x} = -kx. \quad (1.21)$$

Set $m = k = 1$ temporarily to remove constants. Then

$$\ddot{x} = -x. \quad (1.22)$$

Unlike the exponential function, a sine or cosine returns to minus itself after two differentiations. The general solution may be written

$$x(t) = C \cos(t - t_0), \quad (1.23)$$

$$p(t) = \dot{x}(t) = -C \sin(t - t_0). \quad (1.24)$$

The two constants carry the two pieces of initial information. The amplitude C gives the maximum displacement, while t_0 fixes the phase, chosen here so that the velocity vanishes at $t = t_0$. The particle stops only momentarily at the endpoints. It then returns through the origin, where its speed and momentum are largest, and continues oscillating.

Squaring Eqs. (1.23) and (1.24) gives

$$x^2 + p^2 = C^2. \quad (1.25)$$

Thus the phase-space point moves on a circle of radius C . Different initial amplitudes produce different circles; changing t_0 chooses a different starting point on the same circle. Every circle is traversed with the same period 2π in these normalized units, independent of amplitude.

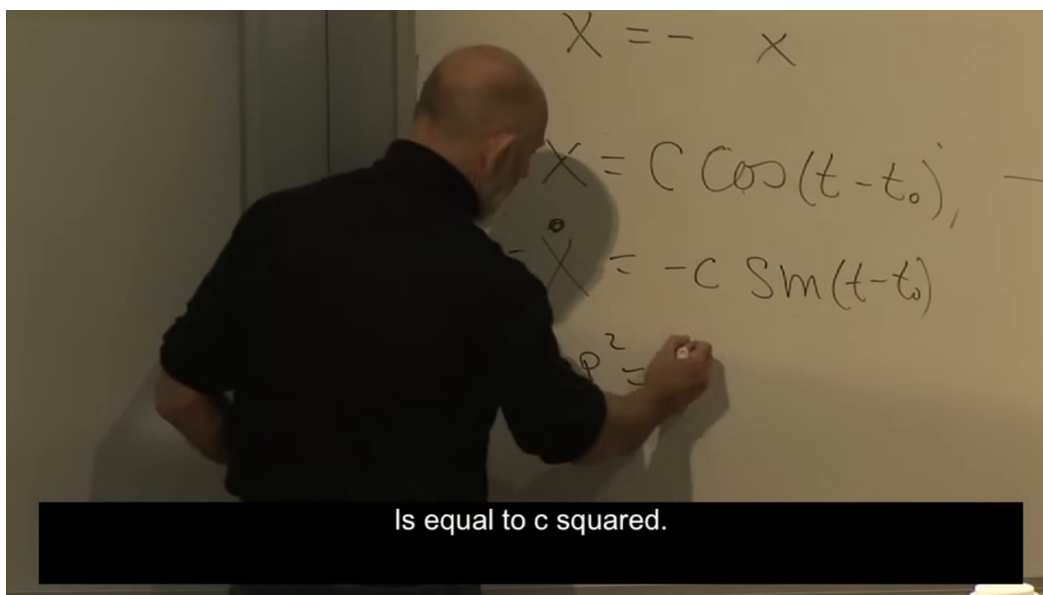


Figure 1.7: The oscillator solution and its circular orbit in the (x, p) phase plane.

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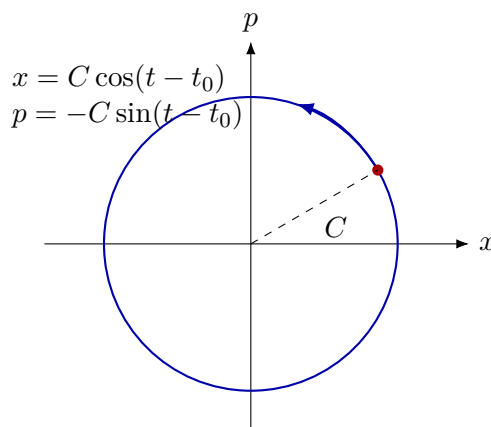


Figure 1.8: A phase-space orbit retains both position and momentum. Its radius labels the motion, while the angle advances uniformly in time.

Projection onto the horizontal axis gives the familiar back-and-forth motion of x ; projection onto the vertical axis gives the oscillation of p . Unlike the Aristotelian spring, distinct circles never collapse into one another. Knowing a phase-space point determines both its future and its past.

For general m and k , the angular frequency is

$$\omega = \sqrt{\frac{k}{m}}, \quad x(t) = C \cos(\omega(t - t_0)), \quad (1.26)$$

and the phase-space orbit becomes an ellipse in the unscaled (x, p) plane. The circular picture is the convenient $m = k = 1$ normalization used in the lecture.

Question from the audience. If the second derivative in the oscillator equation were replaced by a first derivative, would predictability be lost?^a

Response. No. A first-order equation can still predict forward motion from suitable initial data. The stronger issue is whether its flow preserves distinctions well enough to be reversible; the relevant general theorem is deferred until the needed machinery is available.

^aLecture timestamp: 00:45:23–00:46:04.

1.5 Time Reversal

The oscillator picture makes reversibility concrete, but there is also a direct test on the equations. Introduce a reversed time coordinate

$$T = -t. \quad (1.27)$$

Then

$$\frac{d}{dt} = -\frac{d}{dT}, \quad \frac{d^2}{dt^2} = \frac{d^2}{dT^2}. \quad (1.28)$$

A first derivative changes sign under time reversal, while a second derivative does not. Consequently the autonomous Aristotelian equation

$$m\dot{x} = F(x) \quad (1.29)$$

does not retain its form when a movie is run backward. A decaying spring would become a spring that spontaneously departs from equilibrium. By contrast,

$$m\ddot{x} = F(x) \quad (1.30)$$

is unchanged. Reversing a Newtonian movie reverses velocity but leaves acceleration unchanged, so the backward movie is another allowed solution.

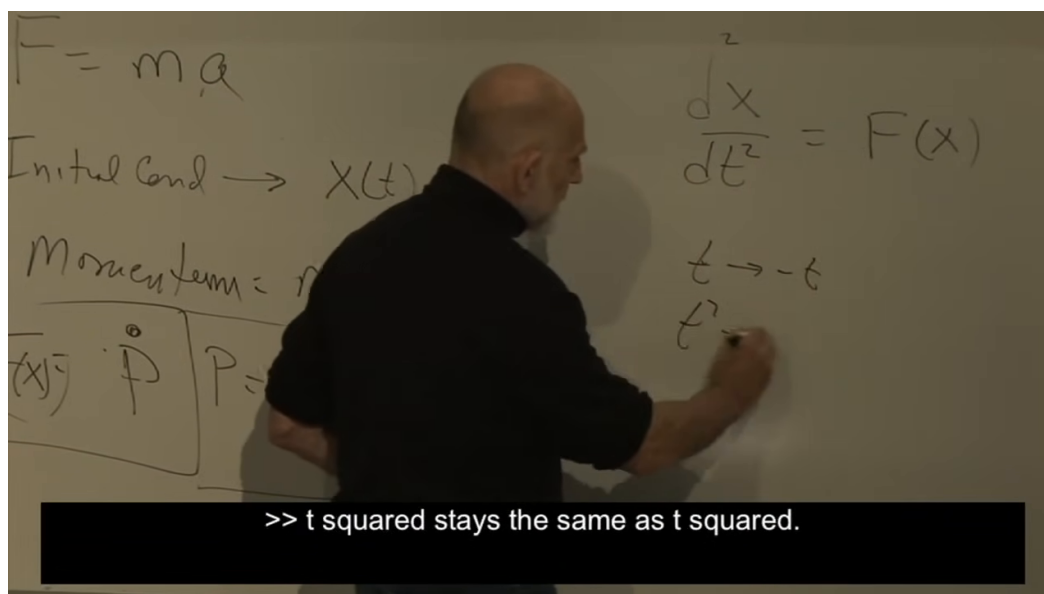


Figure 1.9: The board comparison under $t \mapsto -t$: first derivatives reverse sign, whereas second derivatives do not.

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Question from the audience. Why should reversibility be accepted as a premise about physics in general?^a

Response. It is tested rather than guaranteed by definition. We examine successful laws, identify common experimental features, and elevate those features to principles. The deeper logical order runs in the opposite direction: quantum field theory is more fundamental, and Newtonian mechanics emerges through approximations such as slow motion and heavy objects. A course must build upward from simpler laws, so reversibility is introduced now as a principle whose reach will be tested later.

^aLecture timestamp: 00:46:16–00:48:26.

Question from the audience. Was reversibility itself a problem Newton consciously recognized?^a

Response. The historical answer is uncertain. Newton certainly understood inertia: force is required to change a state of motion, not to sustain uniform motion. Whether he framed that insight explicitly in terms of reversibility is a separate historical question.

^aLecture timestamp: 00:48:28–00:49:02.

1.6 Cycles and Conserved Quantities

The phase-space circles suggest the next idea. A trajectory remains on one circle and never jumps to another. Therefore the circle can be labeled by a number that remains constant along the motion. For the normalized oscillator that number is

$$C^2 = x^2 + p^2. \quad (1.31)$$

Up to a factor of one half, this is the oscillator's energy. Larger circles have larger energy; changing the starting phase does not change it. We shall derive that statement rather than merely read it from the picture, but first consider momentum.

Newton's first law says that an object subject to no net force moves with constant velocity. It is already contained in the second law:

$$\mathbf{F} = 0 \implies \dot{\mathbf{p}} = 0 \implies \mathbf{v} = \text{constant} \quad (1.32)$$

for constant mass. Newton's third law contributes genuinely new information about interactions between particles.

1.6.1 Equal, opposite, and central forces

Let $\mathbf{F}_{ij}$ denote the force on particle i exerted by particle j . The equal-and-opposite part of the third law is

$$\mathbf{F}_{ij} = -\mathbf{F}_{ji}. \quad (1.33)$$

If the two forces pointed in the same direction, an isolated pair could accelerate spontaneously. It would be a propellant-free machine producing motion from internal forces alone. Such behavior is not observed.

For the point-particle forces considered here, the pair also acts along the line joining the particle centers. Equal and opposite forces that were not collinear could form a pure torque and set an isolated rigid pair rotating. The central-force condition prevents that possibility:

$$\mathbf{F}_{ij} \parallel (\mathbf{r}_i - \mathbf{r}_j). \quad (1.34)$$

The equal-and-opposite condition will give conservation of linear momentum. The line-of-centers condition is tied to conservation of angular momentum, which will be studied later.

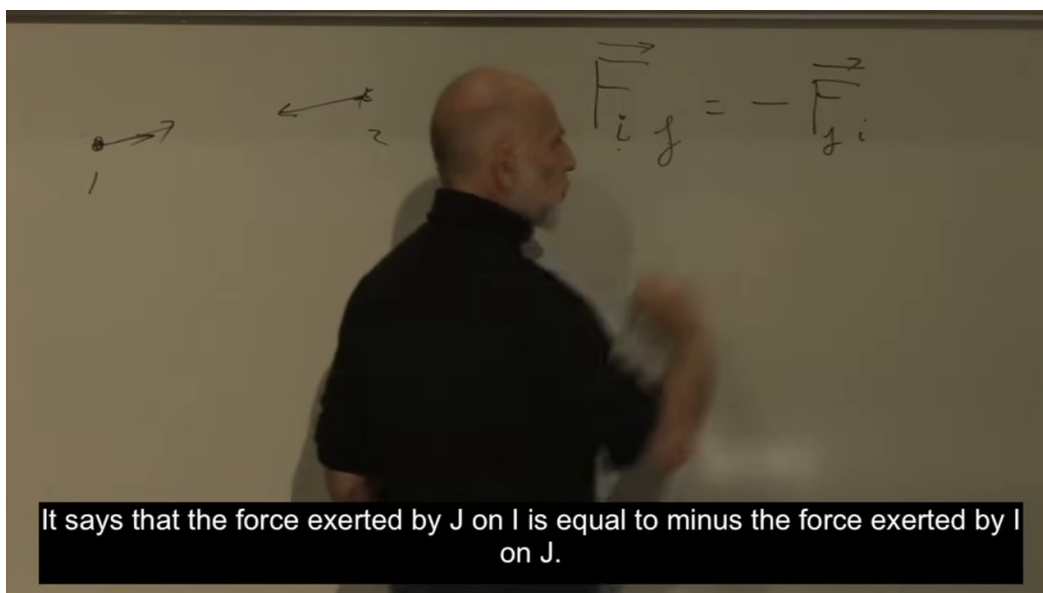


Figure 1.10: The indexed statement of equal and opposite interaction forces.

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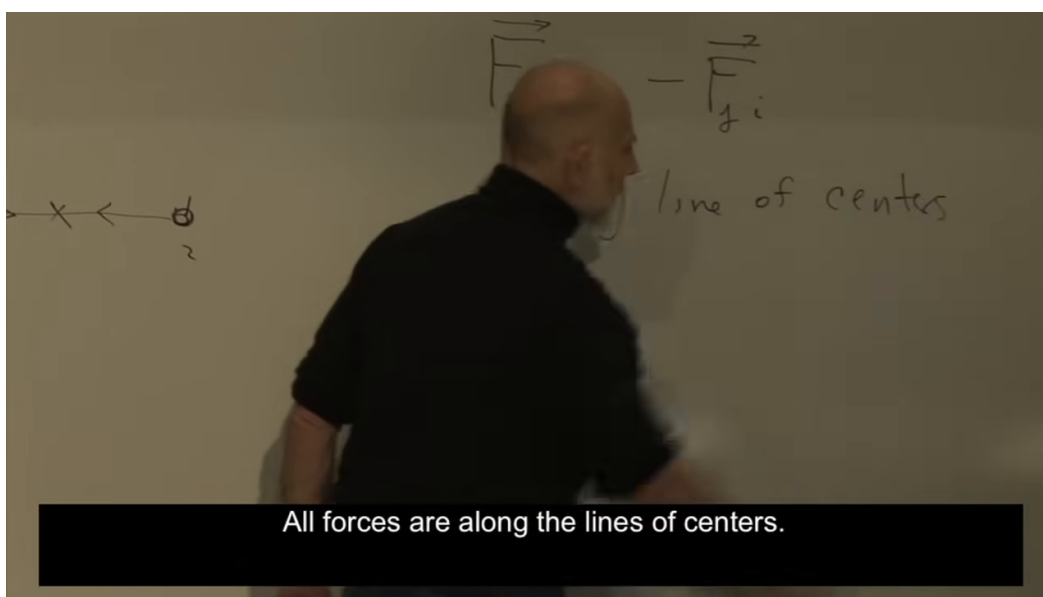


Figure 1.11: The central-force part of Newton's third law: interaction forces lie along the line joining the particles.

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1.6.2 Total momentum

Consider a closed system of N particles. There are no external forces; the only forces are those exerted by the particles on one another. Force additivity and Newton's second law give, for each particle,

$$\dot{\mathbf{p}}_i = m_i \ddot{\mathbf{r}}_i = \sum_{\substack{j=1 \\ j \neq i}}^N \mathbf{F}_{ij}. \quad (1.35)$$

Sum this equation over i :

$$\begin{aligned} \frac{d}{dt} \sum_{i=1}^N \mathbf{p}_i &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \mathbf{F}_{ij} \\ &= \sum_{i < j} (\mathbf{F}_{ij} + \mathbf{F}_{ji}) = 0. \end{aligned} \quad (1.36)$$

Every interaction appears twice, once in each direction, and the pair cancels. Defining

$$\mathbf{P}_{\text{tot}} = \sum_{i=1}^N \mathbf{p}_i, \quad (1.37)$$

we obtain the first conservation law:

$$\frac{d\mathbf{P}_{\text{tot}}}{dt} = 0. \quad (1.38)$$

Historically we have presented momentum conservation as a consequence of Newton's third law. Modern physics often regards momentum conservation as the more fundamental statement: if an isolated two-particle system conserves total momentum, its internal momentum changes must be equal and opposite. For the present development, however, the Newtonian route is the useful one.

1.7 Energy in One Dimension

Take one particle moving on a line under a force that depends only on its position:

$$F = F(x). \quad (1.39)$$

Such a force is called time independent. This does not mean that the force experienced by the moving particle has a constant numerical value. Its value changes as $x(t)$ changes. It means that whenever the particle is at the same position x , it experiences the same force, regardless of the clock time.

In one dimension, any sufficiently regular force function can be integrated. We therefore define a potential energy $V(x)$ by

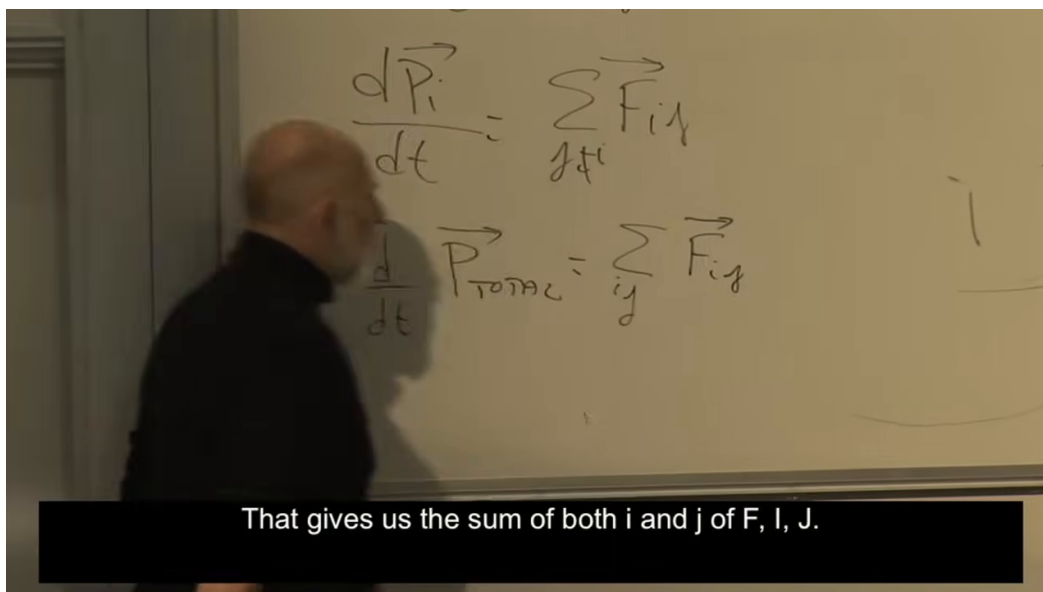


Figure 1.12: Summing the particle equations makes the pairwise forces cancel, leaving the derivative of total momentum equal to zero.

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$$F(x) = -\frac{dV}{dx}, \quad (1.40)$$

or equivalently

$$V(x) = -\int^x F(u) du + V_0. \quad (1.41)$$

The additive constant V_0 has no effect on the force. Picture $V(x)$ as the height of a landscape above the x -axis. The particle is pushed toward lower height, and the force is larger where the slope is steeper. At the top or bottom of a smooth hill the slope vanishes, so the force vanishes there.

Define kinetic, potential, and total energy by

$$T = \frac{1}{2}m\dot{x}^2, \quad E = T + V(x) = \frac{1}{2}m\dot{x}^2 + V(x). \quad (1.42)$$

At this stage energy is simply the name assigned to this candidate conserved quantity. Its physical interpretation is suggested by the landscape: rolling downhill lowers V while increasing speed and therefore T ; climbing uphill does the reverse. Newton's equation proves that the exchange is exact.

Differentiate Eq. (1.42). Since m is constant and V depends on time only through $x(t)$,

$$\begin{aligned} \frac{dE}{dt} &= m\dot{x}\ddot{x} + \frac{dV}{dx}\dot{x} \\ &= \dot{x} \left(m\ddot{x} + \frac{dV}{dx} \right). \end{aligned} \quad (1.43)$$

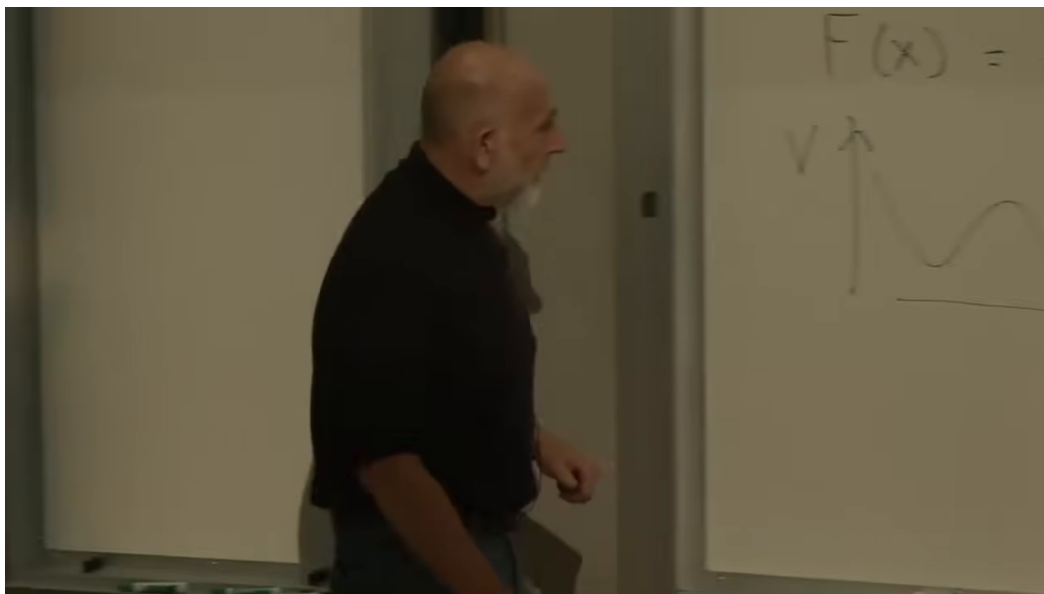


Figure 1.13: Potential energy as a one-dimensional landscape. The force points downhill and has magnitude set by the slope.

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Newton's law and the definition of V say

$$m\ddot{x} = F = -\frac{dV}{dx}. \quad (1.44)$$

The bracket in Eq. (1.43) therefore vanishes:

$$\frac{dE}{dt} = 0. \quad (1.45)$$

No additional dynamical assumption was needed in one dimension beyond a regular position-dependent force, because every such force admits an antiderivative.

1.8 Several Dimensions and Conservative Forces

For one particle with coordinates x_i , Newton's equation has components

$$F_i(\mathbf{x}) = m\ddot{x}_i. \quad (1.46)$$

The force component F_i may depend on all coordinates, not only on x_i . We would like to find one scalar function $V(\mathbf{x})$ such that

$$F_i(\mathbf{x}) = -\frac{\partial V(\mathbf{x})}{\partial x_i}, \quad \mathbf{F} = -\nabla V. \quad (1.47)$$

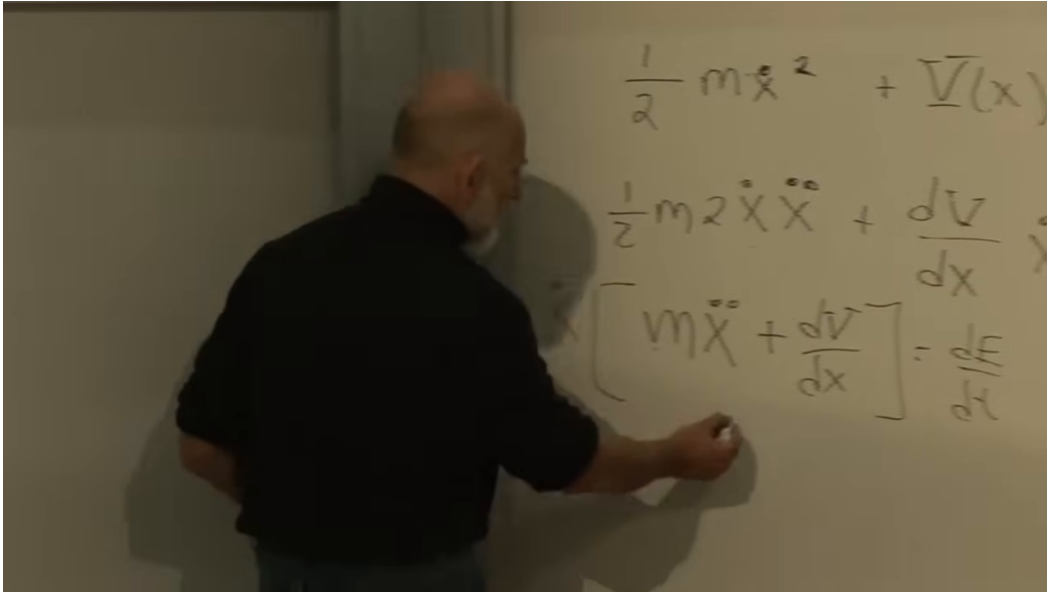


Figure 1.14: The one-dimensional energy derivative factored into velocity times Newton's equation.

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Unlike the one-dimensional construction, this is not automatic. Three arbitrary functions F_1, F_2, F_3 generally contain more independent information than the derivatives of one scalar function. Equivalently, a potential exists only when the force field satisfies compatibility conditions such as equality of the appropriate mixed partial derivatives; in a simply connected region this is expressed by $\nabla \times \mathbf{F} = 0$. We therefore take Eq. (1.47) as a physical postulate for the class of forces under consideration, not as a mathematical identity.

On a contour map, V is constant along each level curve. The gradient is perpendicular to those curves and points toward steepest increase; the force $-\nabla V$ points toward steepest descent.

When Eq. (1.47) holds, define

$$E = \frac{1}{2}m \sum_i \dot{x}_i^2 + V(x_1, \dots, x_n). \quad (1.48)$$

The kinetic term is one half the mass times the squared length of the velocity vector. Differentiating and applying the multivariable chain rule gives

$$\begin{aligned} \frac{dE}{dt} &= m \sum_i \dot{x}_i \ddot{x}_i + \sum_i \frac{\partial V}{\partial x_i} \dot{x}_i \\ &= \sum_i \dot{x}_i \left(m \ddot{x}_i + \frac{\partial V}{\partial x_i} \right) = 0. \end{aligned} \quad (1.49)$$

The proof is exactly the one-dimensional proof decorated with an index, but its premise is stronger: the force components must all arise from one potential.

For N particles in three dimensions, collect all coordinates into one list

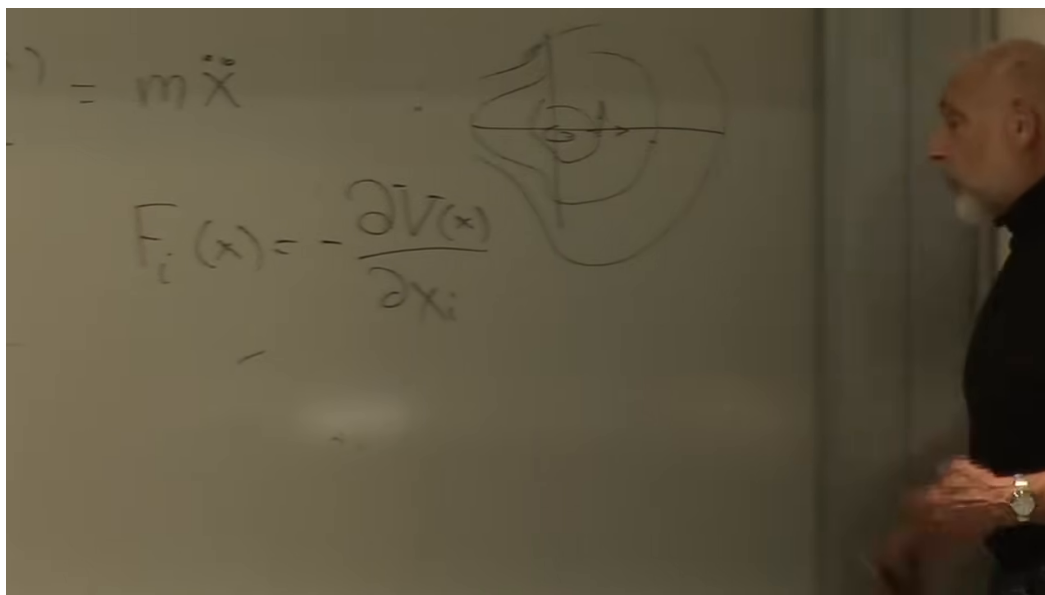


Figure 1.15: The multidimensional force law and a contour sketch of the potential. Force points across level curves toward decreasing V .

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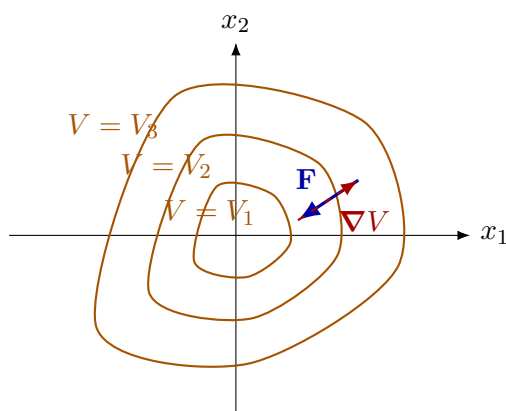


Figure 1.16: Level curves of a scalar potential. The gradient is normal to the contours, and a conservative force points in the opposite direction.

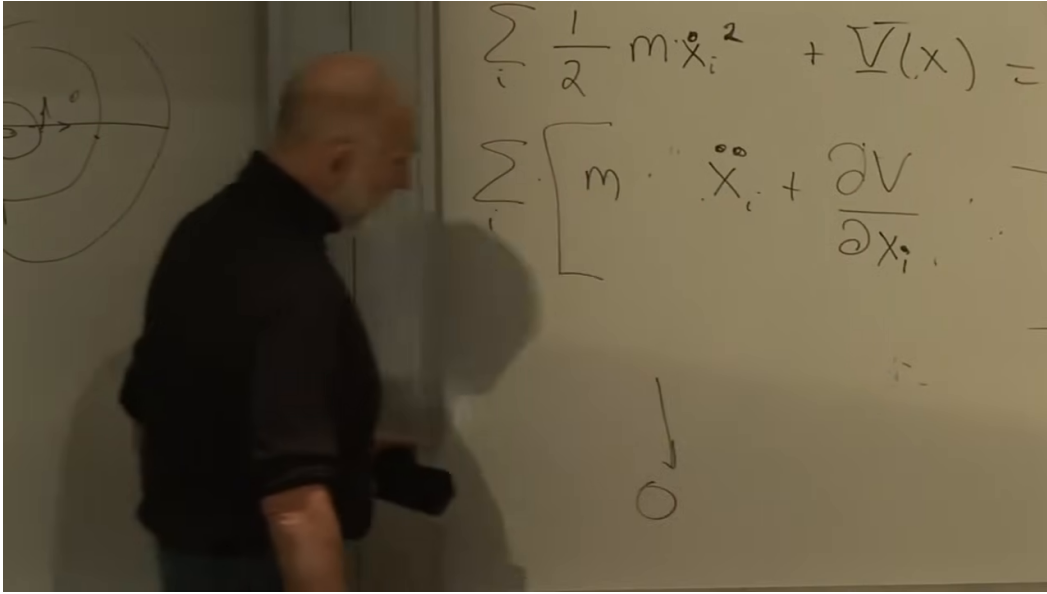


Figure 1.17: The indexed energy derivative. Each bracket vanishes by the corresponding component of Newton's equation.

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$$q_a = (x_1, y_1, z_1, x_2, y_2, z_2, \dots, x_N, y_N, z_N), \quad a = 1, \dots, 3N. \quad (1.50)$$

There are $3N$ coordinate entries but only N particle masses; the same mass occurs in the three entries belonging to a given particle. If

$$F_a = -\frac{\partial V(q_1, \dots, q_{3N})}{\partial q_a}, \quad (1.51)$$

then

$$E = \sum_{A=1}^N \frac{1}{2} m_A |\dot{\mathbf{r}}_A|^2 + V(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad (1.52)$$

is conserved by the same calculation. Forces that arise from a potential are called *conservative* precisely because they support this conservation law; the adjective has no political meaning.

1.9 The Oscillator as an Energy Contour

For the spring force $F = -kx$, Eq. (1.40) gives

$$V(x) = \frac{1}{2} kx^2 + V_0. \quad (1.53)$$

Choosing $V_0 = 0$, the oscillator energy is

$$E = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} kx^2 = \frac{p^2}{2m} + \frac{1}{2} kx^2. \quad (1.54)$$

For $m = k = 1$,

$$2E = p^2 + x^2 = C^2. \quad (1.55)$$

The phase-space circles found earlier are therefore contours of constant energy. For a more general one-dimensional potential, the contours need not be circles, but the motion still lies on curves

$$\frac{p^2}{2m} + V(x) = E \quad (1.56)$$

in the (x, p) plane. This returns us to the idea of the first lecture: when trajectories remain on distinct cycles or orbits, a number can label those orbits and remain unchanged throughout the motion.

We now have the state space, the Newtonian update equations, and the first two conservation laws. The next formulation of mechanics will begin from the principle of least action and lead to Lagrangians, Hamiltonians, angular momentum, and the more general structure behind these examples.

Question from the audience. How closely do the distributed handout and the work on the board correspond, and where are the figures?^a

Response. The handout should follow the lecture closely; if material on the board is absent, the wrong file may have been sent. The figures had not yet been prepared; they would appear, he jokes, when Continuing Studies paid the next check.

^aLecture timestamp: 01:37:56–01:38:54.